Lecture 15 recap

In Lecture 15:

- Introduced eigenvalues and eigenfunctions for $-\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ with periodic BCs
  - They are $\cos(k_n\,x)$ and $\sin(k_n\,x)$ for $k_n = n\,\pi/L$ $(n = 1, 2, \ldots)$
  - and the constant function (we chose $1/2$)
- Showed they are orthogonal with respect to inner product $\langle f, g\rangle = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f\,g\;\mathrm{d}x$

Lecture 16

In Lecture 16, we will:

1. Introduce the property of self-adjointness
2. Define Fourier series
3. Introduce the Euler–Fourier formulae
4. Calculate Fourier series for example functions
5. Motivate convergence study

Generalising real symmetric matrices

Self-adjoint linear transformations

Let T be a linear transformation on a suitable space of functions.* T is self-adjoint if

$$\langle \mathsf{T}f, g\rangle = \langle f, \mathsf{T}g\rangle \ .$$

For a self-adjoint operator T, the eigenvalue problem

$$\mathsf{T}y = \lambda\, y$$

has real-valued eigenvalues λ.

Example:

$\mathsf{T} = -\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ with periodic BCs is self-adjoint under the inner product from last lecture

*The correct choice of function space is crucial, see MAST30031 for a more detailed discussion.

Fourier series

A *Fourier series* on $x \in (-L, L]$ has the form

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty} \Big[a_n \cos(k_n x) + b_n \sin(k_n x) \Big] \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

for some set of coefficients a_n and b_n

- Each term in the Fourier series is called a *mode*
- The coefficients a_n and b_n are also known as the *amplitudes* of the modes
- k_n are *wavenumbers*
- If the independent variable is time we use *angular frequency*, ω_n, instead of k_n

Fourier series

- Every Fourier series thus defined is $2L$-periodic

Fourier series: Coefficients

Suppose that the Fourier series converges for all $x \in (-L, L]$ and that

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \Big[a_n \cos(k_n x) + b_n \sin(k_n x) \Big]$$

The Fourier coefficients are given by the Euler–Fourier formulae

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos(k_n x)\, \mathrm{d}x \qquad \text{and} \qquad b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin(k_n x)\, \mathrm{d}x$$

We call this the Fourier series of $f(x)$ on $x \in (-L, L]$

Exercise 4.3: Find the Fourier series of $f(x) = x^2$ where $x \in (-\pi, \pi]$

First let's calculate the Fourier coefficients.

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 \mathrm{d}x = \frac{1}{\pi}\left[\frac{x^3}{3}\right]_{-\pi}^{\pi} = \frac{2\pi^2}{3}$$

$$\begin{aligned} a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} x^2 \cos nx \,\mathrm{d}x \\ &= \frac{1}{\pi}\left(\left[\frac{x^2}{n}\sin nx\right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi}\frac{2x}{n}\sin nx \,\mathrm{d}x\right) = \frac{2}{\pi}\int_{-\pi}^{\pi}\left(-\frac{x}{n}\sin nx\right)\mathrm{d}x \\ &= \frac{2}{\pi}\left(\left[\frac{x}{n^2}\cos nx\right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi}\frac{1}{n^2}\cos nx \,\mathrm{d}x\right) \\ &= \frac{2}{\pi}\left(\frac{\pi}{n^2}\cos n\pi - \frac{-\pi}{n^2}\cos n\pi\right) = \frac{4(-1)^n}{n^2} \end{aligned}$$

We also have $b_n = 0$ as the integrand is antisymmetric/odd. The Fourier series of x^2 is therefore

$$s(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$$

Exercise 4.4: Find the Fourier series of $f(x) = x$ where $x \in (-\pi, \pi]$

- $a_0 = a_1 = a_2 = \ldots = 0$ as the integrand is antisymmetric
-

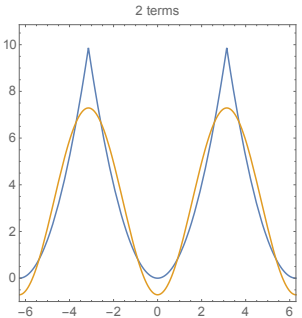
$$
\begin{aligned}
b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx \, \mathrm{d}x \\
&= \frac{1}{\pi} \left(\left[-\frac{x}{n} \cos nx \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} -\frac{1}{n} \sin nx \, \mathrm{d}x \right) \\
&= \frac{1}{\pi} \left(-\frac{\pi}{n} \cos n\pi - \frac{\pi}{n} \cos n\pi \right) = -\frac{2(-1)^n}{n}
\end{aligned}
$$

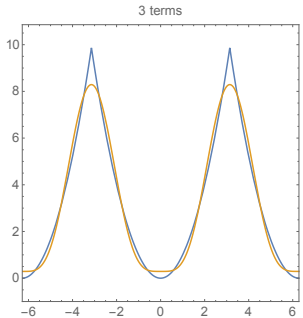
Therefore, the Fourier series of x is

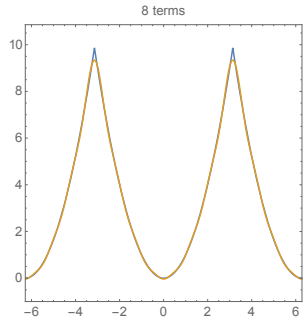
$$
f(x) = -2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin nx
$$

Fourier series of x^2

The Fourier series of $f(x) = x^2$ on $x \in (-\pi, \pi]$ is converging to its *periodic extension*.





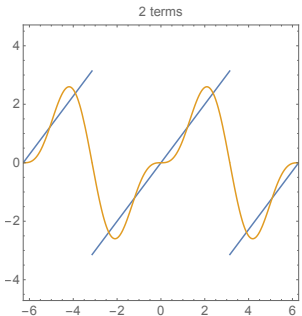


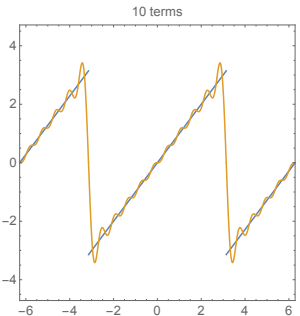
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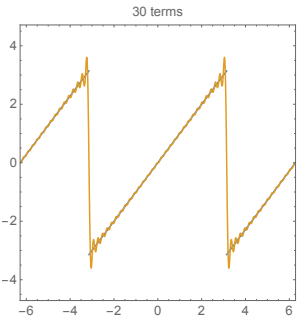
$$\lim_{x \to -\pi^+} f'(x) = -2\pi \neq 2\pi = f'(\pi)$$

Fourier series of x

The Fourier series of $f(x) = x$ on $x \in (-\pi, \pi]$ is converging to its *periodic extension*.







But

$$\lim_{x \to -\pi^+} f(x) = -\pi \neq \pi = f(\pi)$$